

From Industry to Education: Enhancing Chemical Engineering Learning with Operator Training Simulators and Generative AI

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Abstract

Operator Training Simulators (OTS) are commonplace in the chemical engineering industry but often underutilized in universities (Patle et al., 2019). Like a 'flight simulator' for engineers, they are 'digital twins' of real plants, that can run many safety scenarios. In tertiary education OTS offer scalable, active learning environments and authentic assessment, particularly when integrated with Generative AI (GenAI). Our pedagogical design is scaffolded through UTAUT2 (Unified Theory of Acceptance and Use of Technology), offering immersive, industry-aligned, practice-based engineering educational design (Honig et al., 2025), something that is often difficult to do in conventional classroom teaching (Honig et al., 2024).

This presentation on OTS integration will focus on technology-enhanced learning and authentic assessment. It will draw on learnings from the integration of GenAI into the OTS software: while most people think of interacting with GenAI through text-interfaces (like ChatGPT) here students can interact through the game-interface itself (for example opening a valve or if an alarm trips, the GPT 'knows' and can automatically respond).

In response to remote learning challenges presented during COVID-19 lockdowns (Honig et al., 2022), a modified industry-grade OTS (TSC Simulation) was embedded into undergraduate subjects. The simulator, originally designed for professional operator training, was adapted to educational needs by including assessment-focused scenarios and then augmented with a GPT-powered AI teaching assistant. Over four years, it has been used in both second- and third-year core Chemical Engineering subjects, providing students with a unique opportunity to interact with digital twins, analyze process safety incidents, and apply critical thinking in real-time problem-solving.

Using a Design-Based Research framework, the initiative evolved through iterative cycles of student use, feedback, and redesign. Mixed methods evaluation involved pre- and post-use surveys grounded in the UTAUT framework, performance data from quizzes and assignments, and qualitative student feedback. The integration of GenAI was evaluated for usability, performance expectancy, and impact on learning outcomes. Students' comprehension of safety concepts was compared across user groups—with and without chatbot access—using assessments and reflective discussions.

Across cohorts, the OTS was rated highly for its realism and performance benefits, with a Likert average of 4.32 (out of 5) on performance expectancy. The GenAI chatbot, acting as a plant supervisor, facilitated guided root-cause analyses and reflection. Within a limited sample size, students with access to the AI assistant indicated higher quiz performance (67%) than those without (59%). However, effort expectancy for the OTS rated lower, highlighting the complexity of adapting industry-grade software to educational contexts. Improvements were made by redesigning activities to fall within students' Zones of Proximal Development, particularly when supported with new GPT-based adaptive learning assistants, utilizing an agent structure.

This initiative offers a replicable model for incorporating industry technologies and GenAI into curriculum-aligned, scalable assessment formats. It demonstrates how immersive learning tools can address gaps in traditional practicals, support student autonomy, and align with ASE's call for flexible, digitally enhanced, and inclusive educational experiences. We will share initial learnings. Significant broader outcomes have also emerged from the work: as well integrating GPTs into simulators as AI-

assistants for education, GPTs can similarly be integrated into real plants as AI-engineers for process control.

The presentation will outline opportunities for GenAI integration into tertiary education, with a specific focus on integration into simulation-based learning itself (as opposed to interaction through a chat interface).

The presentation will have an interactive component allowing participants to build a customized chatbot through a purpose-built interface for the conference presentation.

References

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